Benchmarking ColVision retrieval models within the `mmore` RAG pipeline

An efficiency and multilingual study
on medical documents

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Abstract

mmore is a multimodal RAG pipeline that offers two ways of retrieving a page: from the text extracted from it, or from its image, encoded by a model of the ColPali family. Until now the second path had accepted only one model; PR #305 added ColQwen2, ColQwen2.5, ColGemma3 and the two ColSmol to it, bringing to six the number of encoders selectable by configuration. Which one to enable remains an open question. We answer it with a frozen pipeline — index, reranker and retrieval contract held constant, only the encoder varies — across two experiments on biomedical corpora, keeping the mmore text path as a reference point. On figure-rich lecture slides, all six encoders beat text retrieval, the best by 28%, but their per-query cost varies by a factor of five and a half without following model size. On five corpora natively written in English, French, Chinese, German and Spanish, no encoder dominates everywhere, and the hybrid mmore text path turns out not to be multilingually robust. We close with a table of choices by deployment context.

1. Introduction

mmore is a modular RAG pipeline for multimodal documents, developed at the LiGHT laboratory of EPFL. Its original chain first turns every document into text — direct extraction when the PDF carries a text layer, OCR otherwise — then indexes that text page by page, combining a dense multilingual embedding and a sparse lexical model, fused at query time. This is an efficient choice as long as the information is carried by words.

It is less so as soon as the page says something its transcription does not. A diagram, a table of results or an annotated lecture slide carry part of their meaning in the figure and in the layout; text extraction keeps their labels and loses the structure. This is the observation formalised by ColPali: rather than transcribing the page, one renders it as an image and encodes it directly into a set of token-level vectors, compared to the query by the late interaction score of ColBERT. The lineage has since been extended to several backbones — ColQwen2 and ColQwen2.5 on Qwen-VL, ColGemma3 on Gemma 3, the two ColSmol on SmolVLM. Each publishes its own scores on ViDoRe and ViDoRe v2, but computed on its authors' software stack: those numbers do not compare directly with one another, and above all they say nothing about how an encoder behaves once inserted into a given pipeline.

The two retrieval paths of mmore. The project therefore offers two ways of indexing the same unit — the page — which do not retain the same signal from it. The **text path** extracts its words (Marker+Surya OCR when the PDF has

no text layer), computes a dense multilingual embedding as well as a sparse SPLADE lexical representation, and fuses the two scores at retrieval time through a weighted ranker: a page is represented there by one vector and a bag of terms. The **colvision path** renders the page as a 200 dpi PNG through PyMuPDF and hands it to a visual encoder, which produces not one vector but several hundred, one per visual token; retrieval issues a first vector top-k then reorders the candidates by late interaction. Both paths return their ranked list through the same *Retriever* contract, which makes them substitutable — and therefore comparable. That is where their cost gap comes from: one compares a single vector per page, the other compares hundreds.

That second path accepted a single encoder, ColPali. PR #305, since merged upstream, added ColQwen2, ColQwen2.5, ColGemma3 and the two ColSmol to it: six encoders are now selectable by plain configuration. The question then becomes a practical one: which to enable, and what does it buy over the text path?

This report answers it across two experiments sharing a frozen pipeline, the first on a corpus of figure-rich lecture slides, the second on scientific literature natively written in five languages. Three questions structure them: whether retrieval quality really separates the six encoders; how their per-query cost is distributed; and what becomes of their performance on documents written in other languages and other scripts.

2. Protocol

2.1. What is held constant

A comparison only means something if the encoder is the only moving part. The following are therefore fixed across all cells — one cell being a single (encoder, corpus) pair: the mmore version (commit `4102f96`, posterior to PR #305), the vector index (Milvus, FLAT, inner product), the reranker (MaxSim on CPU), the depth ($top_k = 10$), the page rendering (200 dpi PNG) and the retrieval contract. On the model side, the precision (`bfloat16`), the batch size (8) and the token vector dimension (128) are shared by all six and follow their published recommendations; none is re-tuned. Each model gets its own Milvus collection. Only the visual encoder and the corpus therefore vary.

2.2. Encoders evaluated

The six encoders (Table 1) cover five distinct backbones and all come from the same version of `colpali-engine`.

Encoder	Backbone	Checkpoint
ColPali v1.3	PaliGemma	`vidore/colpali-v1.3`
ColQwen2 v1.0	Qwen2-VL	`vidore/colqwen2-v1.0`
ColQwen2.5 v0.2	Qwen2.5-VL	`vidore/colqwen2.5-v0.2`
ColGemma3	Gemma-3	`Cognitive-Lab/ColNetraEmbed`
ColSmol-500M	SmolVLM	`vidore/colSmol-500M`
ColSmol-256M	SmolVLM	`vidore/colSmol-256M`

TABLE 1. The six encoders selectable in mmore. ColPali was already there; the five others were added by PR #305.

2.3. The two experiments

Experiment 1: biomedical lecture slides. We adopt ViDoRe v2 (`biomedical_lectures_eng_v2`): 1,016 figure-rich pages and 160 queries written with the page in view then validated by human annotators, with graded, multi-page relevance judgements. Since ViDoRe ships images, we rebuild one PDF per document — one image per page, sized so that re-rendering at 200 dpi reproduces the original pixels — so that the chain runs unchanged. As these pages have no text layer, the text references read the OCR produced by mmore.

Experiment 2: native scientific literature. The second corpus is natively written in each language, never translated. Five languages are evaluated: English, French, simplified Chinese, German and Spanish. Four of them come from PMC-OA, filtered to natively written articles through the NCBI field `<Language>[Language]`; French comes from open-access HAL theses. Each language is capped at the same budget (50 PDFs, 500 pages). The queries — from 432 to 724 depending on the language — are produced by Qwen2.5-32B-Instruct from the page text, and phrased in English whatever the language of the document: retrieval there is cross-lingual.

The choice of generator shapes what the experiment measures. It has to follow an instruction and return valid JSON over thousands of calls, read Chinese, German and Spanish to draw an English question out of them, and fit on the available GPUs; Qwen2.5-32B-Instruct combines all three, whereas domain base models, chief among them Meditron, continue the page text instead of drawing a question from it. It does not see the page: the questions come from the text, which accounts for much of §5. Experiment 1, conversely, rests on questions written with the image in view then validated by humans — the two experiments bracket the problem.

Each corpus ships with a manifest recording, per PDF, its SHA-256 digest, its page count, its language and its source; that digest, and that of the query set, are stored in every result.

2.4. Text references

Two text pipelines run on the same pages, the same queries and the same judgements, with no image signal at all. The **mmore text path** (dense+SPLADE, fusion 0.5, reranker disabled) is what a user who does not enable `colvision` gets: it is the reference that matters. A **dense-only retriever** (`BAAI/bge-m3`, one embedding per page) isolates the lexical stage from the dense one when attributing the gaps.

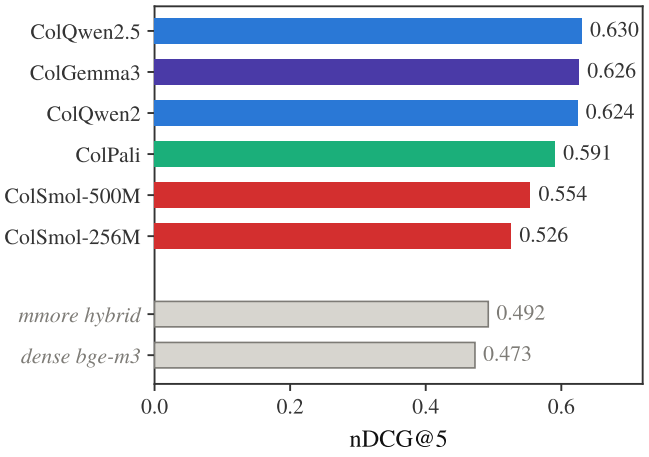


Figure 1. Experiment 1: nDCG@5 of the six encoders. The two grey bars, set apart, are the reference text pipelines.

Encoder	nDCG@5	nDCG@10	MRR	Recall@10	MAP@10
ColQwen2.5	**0.630**	**0.666**	0.717	0.766	**0.573**
ColGemma3	0.626	0.660	0.709	**0.767**	0.564
ColQwen2	0.624	0.657	**0.723**	0.754	0.563
ColPali	0.591	0.627	0.683	0.719	0.536
ColSmol-500M	0.554	0.574	0.629	0.674	0.482
ColSmol-256M	0.526	0.556	0.615	0.661	0.457
mmore, text path	0.492	0.521	0.562	0.625	0.433
dense bge-m3	0.473	0.504	0.560	0.599	0.416

Table 2. Experiment 1: retrieval metrics. The nDCG uses a linear gain, hence is not comparable with published ViDoRe scores.

Metrics and conventions. We report nDCG@5 as the main metric, along with nDCG@10, MRR, Recall@10 and MAP@10. Four conventions govern how they are read. The nDCG uses a **linear** gain ($rel/\log_2(i{+}1)$) and not the exponential gain of the ViDoRe leaderboard: on graded judgements the two diverge, so that our numbers do not compare with published ViDoRe scores. MAP and Recall are capped by $top_k = 10$. Latency, peak VRAM and index size are not instrumented: the only cost measure available is retrieval wall-clock time divided by the number of queries, on a shared GPU. Finally, each cell runs once, without a confidence interval.

3. Experiment 1: lecture slides

3.1. Retrieval quality

On this corpus, enabling the `colvision` path pays off — provided you pick the right encoder (Table 2, Figure 1). All six encoders beat the mmore text path, the best by 28% (0.630 against 0.492), but the spread between them is far wider than their shared margin: from 0.630 for ColQwen2.5 down to 0.526 for ColSmol-256M, whose lead over the text path narrows to 7%. At that model scale the visual path still pays, but barely.

Among the four large encoders, quality barely separates them: ColQwen2.5, ColGemma3 and ColQwen2 lie within 0.006 nDCG@5, with ColPali following at 0.591. The resulting order reproduces the published ViDoRe v2 leaderboard for the models that appear in it, which serves as a consistency check on the ingestion chain.

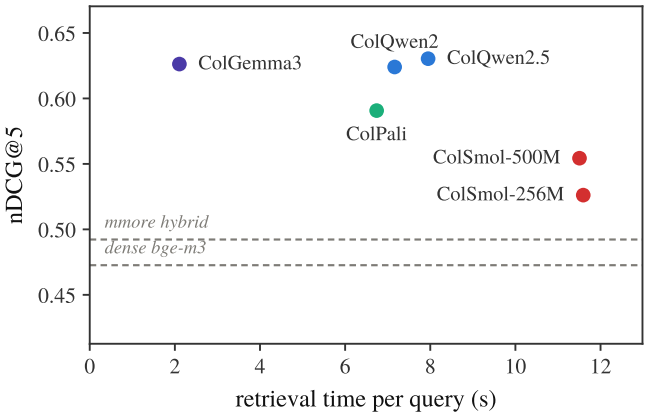


Figure 2. Experiment 1: the same quality, plotted against retrieval time per query. Cost spreads by a factor of five and a half and does not follow model size.

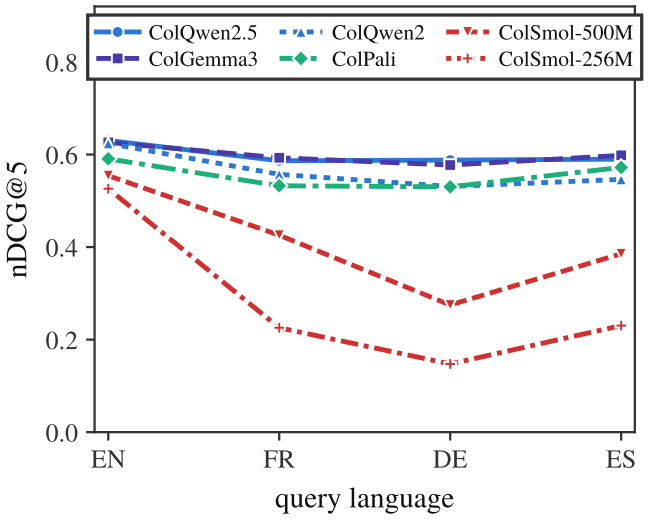


Figure 3. Experiment 1: the same 160 questions, asked in English and in the three languages supplied by the multilingual variant of ViDoRe v2 — Chinese is not among them — on the index built for English: no page is re-encoded, only the query language changes.

3.2. Retrieval cost

It is cost, not quality, that separates the encoders (Figure 2). Retrieval wall-clock time per query ranges from 2.11 s for ColGemma3 to 11.59 s for ColSmol-256M, a factor of 5.5, and it does not follow model size since the smallest encoder is the slowest. The pattern broadly follows the number of vectors a model emits per page, which drives the cost of the MaxSim reranker: ColGemma3 emits only 269, against 766

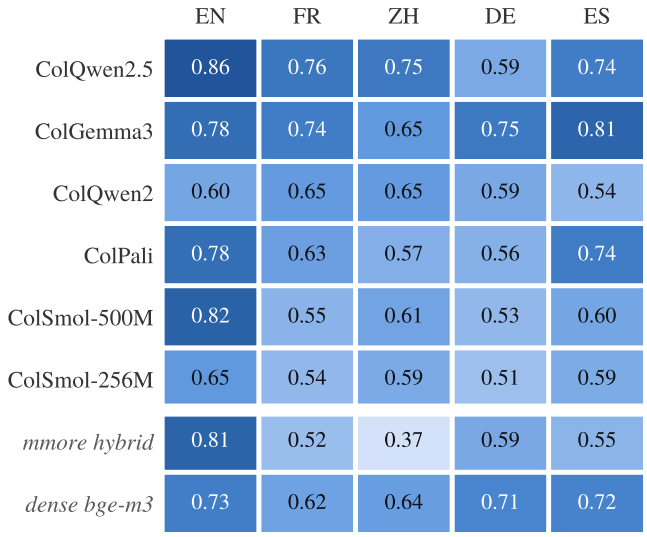


Figure 4. Experiment 2: nDCG@5 on five natively written corpora (50 PDFs / 500 pages per language), queries in English. The last two rows are the text pipelines.

for the Qwen pair, roughly a thousand for the two ColSmol and 1031 for ColPali. The most compact representation is by far the cheapest at query time, and the bulkiest are the slowest. The ordering within the group does not follow all the same — ColPali emits the most vectors yet posts the second-lowest time (6.74 s) — so vector count is not the only term in the cost. Concretely, ColGemma3 reaches leading-group quality for a quarter of the cost of ColQwen2.5 (7.95 s). ColPali, for its part, is the only one of the four large encoders with nothing going for it: it is the weakest in quality (0.591) without being noticeably cheaper, where ColGemma3 pays three times less for its place. These are wall-clock times on shared hardware, so the ratios carry the result and the absolute values do not.

Ingestion times are not reported: the processing configuration enables `skip_already_processed`, so that the measured durations depend on what was already cached and are not comparable across models.

3.3. Robustness to the query language

The multilingual variant of ViDoRe v2 supplies the same 160 questions translated. Replaying them on the index built for English, without re-embedding a single page, isolates the effect of the query language (Figure 3). The four large encoders are stable — losing at most 0.093, ColQwen2.5 going from 0.630 to 0.587–0.590 — whereas the two ColSmol collapse, ColSmol-256M falling from 0.526 in English to 0.147 in German, a factor of 3.6. What separates the models here is the multilingual capacity of their query encoder, and it is also the first property compression sacrifices.

4. Experiment 2: native literature

The second experiment replays the pipeline on five native corpora (Figure 4), with queries in English: retrieval there is cross-lingual. Three observations.

No encoder dominates everywhere. ColQwen2.5 leads in English, French and Chinese; ColGemma3 in German and Spanish, with a gap of 0.158 between the two in German — far beyond the separation observed in experiment 1. The choice of encoder becomes a decision that depends on the language of the corpus, not a global ranking. In each of the five languages, the best encoder stays ahead of the better of the two text pipelines.

Logographic script is not the expected difficulty. One could fear that encoders pre-trained on mostly Latin document images would degrade on Chinese. That is not what we observe: Chinese (0.637 on average over the six encoders) comes out ahead of German for five encoders out of six and reaches the level of French (0.643), and ColQwen2.5, best in English, is also best in Chinese. The hardest language is not the one that changes script but German (0.590), Latin alphabet notwithstanding. The encoders therefore seem to rely more on layout and figure structure, which transfer from one script to another, than on glyph recognition. This ranking of languages nevertheless remains confounded with corpus construction, French being the only one not coming from PMC-OA (§5).

The mmore text path is not multilingually robust. The SPLADE lexical stage helps in English — 0.812 against 0.734 for the dense path alone — and degrades everywhere else, very strongly in Chinese (0.372 against 0.639). The default SPLADE of mmore is an *English* lexical model: on native non-English text, and particularly in CJK, the sparse stage damages the fused ranking instead of improving it. This is a directly actionable result for the project: the text path would benefit from routing on the detected language — hybrid in English, dense-only elsewhere — or from replacing the English SPLADE with a multilingual sparse model.

5. Limitations

Four reservations bound how these numbers should be read.

The queries of experiment 2 are generated by a single model, Qwen2.5-32B-Instruct, from the page text. Two effects follow: they are answerable without the image, which favours a text retriever and makes the measured margin a floor; and the generator shares its backbone with two of the encoders evaluated, ColQwen2 and ColQwen2.5, the latter leading in three languages out of five. With a single generator, we cannot rule out that part of that advantage owes to this kinship.

Context	Recommendation
Default: quality at the best cost	ColGemma3
Maximum quality, cost no object	ColQwen2.5
German or Spanish corpus	ColGemma3
English, French or Chinese corpus	ColQwen2.5
Non-English queries	avoid ColSmol
Memory constraint (not latency)	ColSmol-256M

Table 3. Which encoder to enable according to the deployment context.

One element nevertheless makes it unlikely. Experiment 1 does not use that generator: its questions follow the protocol of ViDoRe v2, phrased from the image of the page then validated by human annotators. The kinship invoked here therefore gets no purchase on it — and ColQwen2.5 still comes out on top. That the same encoder leads in both setups, one of which escapes the objection, suggests that this kinship is not what puts it ahead. Settling the matter for experiment 2 would require regenerating its queries with a model from another family, at equal resources.

The French corpus is asymmetric: under the page budget adopted, it comes down to a few long HAL theses against some fifty short articles for the other languages, so that its scores are confounded with low document diversity and do not read as a language effect.

Each cell runs once. The gaps read as point estimates, and in particular the 0.006 nDCG@5 that separate the top three encoders (Table 2) do not constitute a ranking.

Finally, the absolute values are conditioned by the implementation fixed in §2.1 and by a single GPU class under heterogeneous placement. Another RAG stack or a tighter memory constraint may reorder the models.

6. Conclusion

Enabling the `colvision` path of mmore clearly improves retrieval on documents where the page carries more than its text: on biomedical lecture slides, all six encoders beat the text path, the best by 28%. On scientific literature in five languages, the best encoder stays ahead in every language, but it is not the same one depending on the language.

The choice therefore does not play out on raw quality (Table 3): the top three encoders lie within 0.006 nDCG@5 in experiment 1, but within a factor of four on per-query cost — five and a half across all six. ColGemma3 is the reasonable default, ColQwen2.5 the maximum-quality choice, ColPali the only one hard to recommend, and the two ColSmol are justified only under a strong memory constraint — never under a latency budget, where they are the most expensive of the six, and not at all if the queries are not in English.

One last result concerns mmore independently of ColVision: its hybrid text path is not multilingually robust, its English lexical stage degrading the ranking as soon as the documents are no longer English.

The cells of both experiments and the text references are published as traceable records, along with the corpus manifests, the query sets and the code that produces them (benchmark repository).

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